



Technical communiqué

Matrix-weighted consensus and its applications[☆]Minh Hoang Trinh^a, Chuong Van Nguyen^a, Young-Hun Lim^b, Hyo-Sung Ahn^{a,*}^a School of Mechanical Engineering, Gwangju Institute of Science and Technology (GIST), Gwangju, Republic of Korea^b Department of Electronic Engineering, Gyeongnam National University of Science and Technology, Jinju, Republic of Korea

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ABSTRACT

This paper proposes the matrix-weighted consensus algorithm, which is a generalization of the consensus algorithm. Given a networked dynamical system where the interconnections between agents are weighted by nonnegative definite matrices, it is shown that consensus and clustering phenomena naturally exist. We examine algebraic and algebraic graph conditions for achieving a consensus.

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1. Introduction

Given a system of n single-integrator agents whose interconnections between agents are described by a weighted undirected graph $\mathcal{G}$, the consensus algorithm (Olfati-Saber & Murray, 2004) is defined as:

$$\dot{\mathbf{x}}_i = \sum_{j=1}^n a_{ij}(\mathbf{x}_j - \mathbf{x}_i), \quad \forall i = 1, \dots, n, \quad (1)$$

where $\mathbf{x}_i, \mathbf{x}_j \in \mathbb{R}^d$ are the state vectors of agents i and j , and a_{ij} is a positive scalar (or zero) if i and j are connected (or disconnected, respectively). It is well-known that under the consensus protocol (1), an average consensus is globally achieved if and only if $\mathcal{G}$ is connected (Olfati-Saber & Murray, 2004). This paper generalizes the consensus algorithm (1) by using matrix weight $\mathbf{A}_{ij}$ instead of the scalar weight a_{ij} to describe the interconnection between two agents i and j . A matrix weight could be a positive definite matrix (full connection), a positive semidefinite matrix (partial connection), or a zero matrix (no direct connection). Thus, the matrix-weight consensus covers a larger set of problems in multi-agent systems.

In the literature, matrix-weights have been often used in networked systems to describe the interconnections between agents. For example, the author of Tuna (2016) used matrix weights to describe interconnections between coupled oscillators and provided

conditions for synchronization. The concept of deviated cyclic pursuit introduced in Ramirez, Pavone, Frazzoli, and Miller (2009) and orientation estimation in Lee and Ahn (2016) can be considered as consensus protocols with rotation matrix weights (Ahn, Trinh, & Lee, 2017). Also, the bearing-based network localization setup in Zhao and Zelazo (2016) can be considered as a special case of the matrix-weighted consensus protocol proposed in this paper. In the context of social networks, when a group of people are discussing multiple topics, matrix weights can be used to describe the logical inter-dependency of the topics (Friedkin, Proskurnikov, Tempo, & Parsegov, 2016).

This paper studies the *matrix-weighted consensus algorithm* with fixed undirected graphs. First, we define several terminologies and propose the matrix-weighted consensus algorithm. Second, a necessary and sufficient condition for globally exponentially reaching an average consensus based on the nullspace of the matrix-weighted Laplacian is provided. Third, due to the existence of semidefinite matrix weights, clustered consensus happens even when the graph is connected. We examine the algebraic graph conditions for consensus and clustered consensus. Finally, an algorithm to determine all clusters in the network is provided. The algorithm initially partitions the graph into a set of clusters associated with the positive trees in the graph. If two clusters satisfy several algebraic conditions on their connections, they will be merged together at each iteration. Consequently, the algorithm gradually reduces the number of clusters in the graph, and it ends when no two clusters can be further merged together. If there is a cluster containing all vertices in the graph, a consensus is globally achieved.

2. Preliminaries and problem formulation

A fixed undirected graph with matrix weights is denoted by $\mathcal{G}$. The graph $\mathcal{G}$ is characterized by a triple $(\mathcal{V}, \mathcal{E}, \mathcal{A})$. Here, $\mathcal{V} = \{1, \dots, n\}$ denotes the set of $|\mathcal{V}| = n$ vertices, $\mathcal{E} = \{e_{ij} = (i, j) \mid i, j \in$

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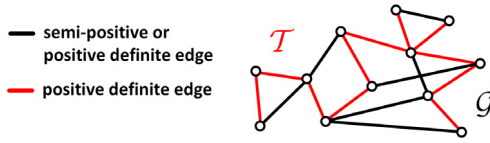


Fig. 1. $\mathcal{T}$ is a positive spanning tree of $\mathcal{G}$.

$\mathcal{V}$, $i \neq j$ denotes the set of $|\mathcal{E}| = m$ edges, and $\mathcal{A} = \{\mathbf{A}_{ij} \in \mathbb{R}^{d \times d} | (i, j) \in \mathcal{E}, \mathbf{A}_{ij} = \mathbf{A}_{ji}^T \geq 0\}$ denotes the set of matrix weights. The dimension d ($d \geq 1$) of the matrix weights in $\mathcal{A}$ depends on the problem. If $d = 1$, $\mathcal{G}$ is an usual undirected scalar-weighted graph. Depending on the matrix weights, the interconnections between vertices in $\mathcal{G}$ are classified into two types. If the weight matrix $\mathbf{A}_{ij}$ corresponding to an edge (i, j) is positive definite (positive semi-definite), we say that (i, j) is a positive (semi-positive) definite edge and two vertices i and j are connected via a positive (semipositive) definite edge. If i and j are disconnected, $\mathbf{A}_{ij} = \mathbf{0}$. We also assume that the interconnections between any two vertices are symmetric, i.e., $\mathbf{A}_{ij} = \mathbf{A}_{ji}$, $\forall (i, j) \in \mathcal{E}$. A path is a sequence of vertices in $\mathcal{G}$, denoted by $\mathcal{P} = i_1 i_2 \dots i_l$, such that each edge (i_k, i_{k+1}) , $k = 1, \dots, l-1$, is a positive/semipositive definite edge. The graph $\mathcal{G}$ is called positive semiconnected if and only if there exists a path between any two vertices in $\mathcal{G}$. Otherwise, $\mathcal{G}$ is disconnected.

Definition 1 (Positive Path). A positive path is a sequence of vertices in $\mathcal{G}$, denoted by $\mathcal{P} = i_1 i_2 \dots i_l$, such that each edge (i_k, i_{k+1}) , $k = 1, \dots, l-1$, is a positive definite edge.

A tree is a graph (containing at least one vertex) in which any two vertices are connected by exactly one path.

Definition 2 (Positive Tree). A positive tree $\mathcal{T}$ is a tree contained in $\mathcal{G}$ having all positive connections. Equivalently, for all $i, j \in \mathcal{V}$, there exists a positive path in $\mathcal{T}$ connecting i and j .

Definition 3 (Positive Spanning Tree). A positive spanning tree $\mathcal{T}$ of $\mathcal{G}$ is a positive tree containing all vertices in $\mathcal{V}$.

Note that a tree of k vertices ($k \geq 1$) contains exactly $k - 1$ edges. Thus, a positive spanning tree of $\mathcal{G}$ contains exactly $n - 1$ positive edges (see Fig. 1). Next, we define several algebraic structures corresponding to the matrix weighted graph $\mathcal{G}$. The *matrix-weighted adjacency matrix* of $\mathcal{G}$ is defined as $\mathbf{A} = [\mathbf{A}_{ij}]_{i,j=1,\dots,n}$. Since $\mathcal{G}$ is undirected and $\mathbf{A}_{ij} = \mathbf{A}_{ji}$, the matrix $\mathbf{A}$ is symmetric. For each vertex i , the neighbor set of i is defined as $\mathcal{N}_i = \{j \in \mathcal{V} | e_{ij} \in \mathcal{E}\}$. Let $\mathbf{D}_i = \sum_{j \in \mathcal{N}_i} \mathbf{A}_{ij}$. We define $\mathbf{D} = \text{blkdiag}(\mathbf{D}_i)$ as the degree matrix of the graph $\mathcal{G}$. The *matrix-weighted Laplacian* is defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$ (Tuna, 2016).

Consider an arbitrary index of the edges of $\mathcal{G}$, we can write the edge set and the matrix-weight set as $\mathcal{E} = \{e_{kij}\}_{k=1,\dots,m}$ and $\mathcal{A} = \{\mathbf{A}_{kij}\}_{k=1,\dots,m}$, correspondingly.¹ Let $\mathbf{H} \in \mathbb{R}^{m \times n}$ be the incidence matrix corresponding to an arbitrary orientation of the edges in $\mathcal{E}$, and let $\bar{\mathbf{H}} = \mathbf{H} \otimes \mathbf{I}_d$, where $\otimes$ denotes the Kronecker product and $\mathbf{I}_d$ is the $d \times d$ identity matrix. We have the following lemma whose proof can be found in Trinh and Ahn (2017).

Lemma 1. The matrix-weighted Laplacian can be written in the following form $\mathbf{L} = \bar{\mathbf{H}}^T \text{blkdiag}(\mathbf{A}_k) \bar{\mathbf{H}}$.

Consider a system consisting of n single integrator agents. Suppose an agent i in the system has a state vector $\mathbf{x}_i = [x_{i1}, \dots, x_{id}]^T \in \mathbb{R}^d$, where $d \geq 1$. Denote $\mathbf{x} = [\mathbf{x}_1^T, \dots, \mathbf{x}_n^T]^T \in \mathbb{R}^{dn}$. The

agents update their states under the following matrix-weighted consensus protocol:

$$\dot{\mathbf{x}}_i = \sum_{j \in \mathcal{N}_i} \mathbf{A}_{ij} (\mathbf{x}_j - \mathbf{x}_i), \quad \forall i = 1, \dots, n. \quad (2)$$

The n -agent system's dynamics can be expressed in the following matrix form:

$$\dot{\mathbf{x}} = -\mathbf{L}\mathbf{x}, \quad (3)$$

where $\mathbf{L}$ is the matrix-weighted Laplacian.

Definition 4 (Consensus). The n -agent system is said to achieve a consensus if and only if $\mathbf{x}_i = \mathbf{x}_j$, for all $i, j \in \mathcal{V}$, $i \neq j$.

Define $\mathcal{R} = \text{Range}\{\mathbf{1}_n \otimes \mathbf{I}_d\}$ as the consensus space. A consensus of the n -agent system is globally/locally asymptotically achieved if and only if $\mathbf{x}$ globally/locally asymptotically approaches $\mathcal{R}$. Although consensus is an important objective, in some applications, the agents' states are desired to converge to some different values. A partition of $\mathcal{V}$ is given by $\mathcal{C}_1, \dots, \mathcal{C}_l$ ($1 \leq l \leq n$) satisfying two properties: (i) $\mathcal{C}_i \cap \mathcal{C}_j = \emptyset$, for $i \neq j$, and (ii) $\bigcup_{k=1}^l \mathcal{C}_k = \mathcal{V}$.

Definition 5 (Cluster Consensus). The n -agent system is said to achieve a cluster consensus if there exists a partition $\mathcal{C}_1, \dots, \mathcal{C}_l$, such that all agents belonging to the same partition achieve consensus, while for any two agents i and j belonging to two different partitions, $\mathbf{x}_i \neq \mathbf{x}_j$. Each $\mathcal{C}_i$, $i = 1, \dots, l$, is referred to as a cluster.

3. Algebraic condition for reaching a consensus

This section aims to find an algebraic condition of the matrix-weighted Laplacian for reaching a consensus.

Lemma 2. The matrix-weighted Laplacian $\mathbf{L}$ is symmetric, positive semidefinite, and has the nullspace $\mathcal{N}(\mathbf{L}) = \text{span}\{\mathcal{R}, \{\mathbf{v} = [\mathbf{v}_1^T, \dots, \mathbf{v}_n^T]^T \in \mathbb{R}^{dn} | (\mathbf{v}_i - \mathbf{v}_j) \in \mathcal{N}(\mathbf{A}_{ij}), \forall (i, j) \in \mathcal{E}\}\}$.

Proof. From Lemma 1, we can write

$$\mathbf{L} = \bar{\mathbf{H}}^T \text{blkdiag}(\mathbf{A}_k) \bar{\mathbf{H}} = \mathbf{M}^T \mathbf{M}, \quad (4)$$

where $\mathbf{M} = \text{blkdiag}(\mathbf{A}_k^{1/2}) \bar{\mathbf{H}}$. Eq. (4) shows that $\mathbf{L}$ is positive semidefinite, and $\mathcal{N}(\mathbf{L}) = \mathcal{N}(\mathbf{M}^T \mathbf{M}) = \mathcal{N}(\mathbf{M})$. As a result, $\mathcal{N}(\mathbf{L}) = \mathcal{N}(\mathbf{M}) \supseteq \mathcal{N}(\bar{\mathbf{H}}) \supseteq \mathcal{R}$. Consider $\mathbf{v} = [\mathbf{v}_1^T, \dots, \mathbf{v}_n^T]^T \notin \mathcal{R}$ such that $\mathbf{L}\mathbf{v} = \mathbf{0}$. It follows $\mathbf{v}^T \mathbf{L}\mathbf{v} = 0$, or i.e., $\sum_{(i,j) \in \mathcal{E}} (\mathbf{v}_i - \mathbf{v}_j)^T \mathbf{A}_{ij} (\mathbf{v}_i - \mathbf{v}_j) = 0$, which implies that $(\mathbf{v}_i - \mathbf{v}_j) \in \mathcal{N}(\mathbf{A}_{ij})$, for all $(i, j) \in \mathcal{E}$. ■

Remark 1. According to Lemma 2, $\dim(\mathcal{N}(\mathbf{L})) \geq \dim(\mathcal{R})$. The matrix-weighted Laplacian $\mathbf{L}$ has at least d zero eigenvalues. Let $\{\lambda_i\}_{i=1,\dots,dn}$ be $\mathbf{L}$'s eigenspectra, we have $0 = \lambda_1 = \dots = \lambda_d \leq \lambda_{d+1} \leq \dots \leq \lambda_{dn}$.

Lemma 3. Under the matrix-weighted consensus protocol (2), the average $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ is invariant.

Proof. We can write $\bar{\mathbf{x}} = \frac{1}{n} (\mathbf{1}_n^T \otimes \mathbf{I}_d) \mathbf{x}$. Taking the derivative of $\bar{\mathbf{x}}$ along the trajectory of (3) yields

$$\dot{\bar{\mathbf{x}}} = \frac{1}{n} (\mathbf{1}_n^T \otimes \mathbf{I}_d) \dot{\mathbf{x}} = -\frac{1}{n} (\mathbf{1}_n^T \otimes \mathbf{I}_d) \mathbf{L} \mathbf{x}.$$

Since $\mathbf{L}$ is symmetric, if $\mathbf{v} \in \mathcal{N}(\mathbf{L})$, then $\mathbf{v}^T$ belongs to the left nullspace of $\mathbf{L}$. Thus, $\dot{\bar{\mathbf{x}}} = -\frac{1}{n} (\mathbf{1}_n^T \otimes \mathbf{I}_d) \mathbf{L} \mathbf{x} = \mathbf{0}$, or i.e. $\bar{\mathbf{x}}$ is invariant. ■

Theorem 1 (Stability). Assume that $\mathcal{G}$ is positive semiconnected. Any trajectory of (3) globally asymptotically approaches the set $\mathcal{N}(\mathbf{L})$.

¹ If it is unimportant to specify the end-vertices explicitly, we will dropout the subscript ij and write e_k and $\mathbf{A}_k$ without ambiguity.

Proof. Consider the potential function $V = \frac{1}{2}\|\mathbf{x}\|^2$, which is positive definite, radially unbounded, and continuously differentiable. Further, $\dot{V} = -\mathbf{x}^T \mathbf{L} \mathbf{x} = -\sum_{(i,j) \in \mathcal{E}(\mathcal{G})} (\mathbf{x}_i - \mathbf{x}_j)^T \mathbf{A}_{ij} (\mathbf{x}_i - \mathbf{x}_j) \leq 0$. It follows $\|\mathbf{x}(t)\| \leq \|\mathbf{x}(0)\|$, or i.e., $\mathbf{x}(t)$ is bounded. Further, $\dot{V}$ is negative semidefinite and $\dot{V} = 0$ if and only if $\mathbf{x} \in \mathcal{N}(\mathbf{L})$. Based on LaSalle's invariance principle, any trajectory of (3) globally asymptotically approaches $\mathcal{N}(\mathbf{L})$. ■

Lemma 4. If $\mathcal{N}(\mathbf{L}) = \mathcal{R}$, the system (3) has a unique equilibrium point $\mathbf{x}^* = \mathbf{1}_n \otimes \bar{\mathbf{x}}$.

Proof. We prove that $\mathbf{1}_n \otimes \bar{\mathbf{x}}$ is the unique equilibrium of (3) if $\mathcal{N}(\mathbf{L}) = \mathcal{R}$ by contradiction. Let $\{\mathbf{e}_i\}_{i=1,\dots,d}$ be a basis of $\mathbb{R}^d$, where $\mathbf{e}_i = [0, \dots, 1, \dots, 0]^T$ is a vector with all zero entries except for an 1 on the i th row. Suppose that there exists $\mathbf{x}' \in \mathcal{R}$ such that $\mathbf{x}' \neq \mathbf{x}^*$. We can write $\mathbf{x}' = \sum_{i=1}^d \bar{x}'_i (\mathbf{1}_n \otimes \mathbf{e}_i) = \mathbf{1}_n \otimes \bar{\mathbf{x}}' = (\mathbf{1}_n \otimes \mathbf{I}_d) \bar{\mathbf{x}}'$, where $\bar{\mathbf{x}}' = [\bar{x}'_1, \dots, \bar{x}'_d]^T$. It follows from Lemma 3 that

$$\begin{aligned} \bar{\mathbf{x}} &= \frac{1}{n} (\mathbf{1}_n^T \otimes \mathbf{I}_d) \mathbf{x}' = \frac{1}{n} (\mathbf{1}_n^T \otimes \mathbf{I}_d) (\mathbf{1}_n \otimes \mathbf{I}_d) \bar{\mathbf{x}}' \\ &= \frac{1}{n} (\mathbf{1}_n^T \mathbf{1}_n \otimes \mathbf{I}_d) \bar{\mathbf{x}}' = \frac{1}{n} (n \otimes \mathbf{I}_d) \bar{\mathbf{x}}' = \bar{\mathbf{x}}'. \end{aligned}$$

As a result, $\mathbf{x}' = \mathbf{1}_n \otimes \bar{\mathbf{x}} = \mathbf{x}^*$, which is a contradiction. Thus, $\mathbf{x}^* = \mathbf{1}_n \otimes \bar{\mathbf{x}}$ is the unique equilibrium of (3). ■

Theorem 2 (Average Consensus). The system (3) globally exponentially converges to the system's average $\mathbf{x}^* = \mathbf{1}_n \otimes \bar{\mathbf{x}}$ if and only if $\mathcal{N}(\mathbf{L}) = \mathcal{R}$.

Proof (Necessity). We prove by contradiction. Assume that (3) globally asymptotically converges to $\mathbf{x}^* = \mathbf{1}_n \otimes \bar{\mathbf{x}}$ but $\mathcal{N}(\mathbf{L}) \neq \mathcal{R}$. From Lemma 2, there exists $\mathbf{x}' \in \mathbb{R}^{dn}$ such that $\mathbf{L} \mathbf{x}' = \mathbf{0}$ and $\mathbf{x}' \notin \mathcal{R}$. Thus, $\mathbf{x} = \mathbf{x}'$ is also an equilibrium point of (3), and any trajectory with $\mathbf{x}(0) = \mathbf{x}'$ stays at $\mathbf{x}'$ for all $t \geq 0$. Thus, $\mathbf{x}^*$ is not globally asymptotically stable, which contradicts the assumption. (Sufficiency): Suppose that $\mathcal{N}(\mathbf{L}) = \mathcal{R}$. Following the proof of Theorem 1, any trajectory of (3) converges to $\mathcal{R}$. From Lemma 4, $\mathbf{x}^* = \mathbf{1}_n \otimes \bar{\mathbf{x}} \in \mathcal{N}(\mathbf{L})$ is the unique equilibrium point of (3). Consider the Lyapunov function $V = \frac{1}{2} \delta^T \delta$, where $\delta = \mathbf{x} - \mathbf{1}_n \otimes \bar{\mathbf{x}}$ is the disagreement vector. Then, V is positive definite, radially unbounded, continuously differentiable, and $\dot{V} = \delta^T \dot{\delta} = -\delta^T \mathbf{L} \mathbf{x} = -\delta^T \mathbf{L} \delta \leq 0$, where in the third equality, we have used the fact that $\mathbf{L} \delta = \mathbf{L} \mathbf{x} - \mathbf{L} (\mathbf{1}_n \otimes \bar{\mathbf{x}}) = \mathbf{L} \mathbf{x} - \mathbf{L} (\mathbf{1}_n \otimes \mathbf{I}_d) \bar{\mathbf{x}} = \mathbf{L} \mathbf{x}$. Moreover, $\delta \perp \mathcal{R}$ since $(\mathbf{1}_n \otimes \mathbf{I}_d)^T \delta = (\mathbf{1}_n \otimes \mathbf{I}_d)^T \mathbf{x} - (\mathbf{1}_n^T \mathbf{1}_n \otimes \mathbf{I}_d) \bar{\mathbf{x}} = n \bar{\mathbf{x}} - n \bar{\mathbf{x}} = \mathbf{0}$. Hence,

$$\dot{V} = -\delta^T \mathbf{L} \delta \leq -\lambda_{d+1}(\mathbf{L}) \delta^T \delta = -\alpha V \leq 0, \quad (5)$$

where $\alpha = 2\lambda_{d+1}(\mathbf{L}) > 0$. Further, $\dot{V} = 0$ if and only if $\delta = \mathbf{0}$, or $\mathbf{x} = \mathbf{x}^* = \mathbf{1}_n \otimes \bar{\mathbf{x}}$. Thus, $\mathbf{x}^*$ is a globally exponentially stable equilibrium of (3). ■

Remark 2. Eq. (5) shows that λ_{d+1} , the smallest positive eigenvalue of $\mathbf{L}$, determines the convergence rate of the matrix-weighted consensus protocol (3). Thus, λ_{d+1} is a performance index of the network, and this index is analogous to the algebraic connectivity of $\mathcal{G}$ in the scalar-weighted consensus algorithm (Mesbahi & Egerstedt, 2010; Olfati-Saber, Fax, & Murray, 2007).

Corollary 1. Under the consensus protocol (2), if $\mathbf{A}_{ij} > 0$ for all $(i, j) \in \mathcal{E}$, all agents globally exponentially achieve a consensus.

Proof. Since $\mathbf{A}_k > 0, \forall k = 1, \dots, m$, we have $\mathcal{N}(\mathbf{L}) = \mathcal{N}(\mathbf{M}) = \mathcal{N}(\text{diag}(\mathbf{A}_k^{1/2}) \bar{\mathbf{H}}) = \mathcal{N}(\bar{\mathbf{H}}) = \mathcal{R}$. Thus, the claim follows immediately from Theorem 2. ■

4. Theory of consensus and clustered consensus phenomena

In the previous section, Theorem 2 provides an algebraic condition for reaching a consensus. However, that condition requires finding the nullspace of $\mathbf{L}$. Further, the requirement for achieving consensus in Corollary 1 could be conservative. This section aims to find conditions for consensus/cluster consensus related with the matrix-weighted graph $\mathcal{G}$.

Lemma 5. If $\mathcal{G}$ has a positive spanning tree $\mathcal{T}$, the n -agent system globally exponentially achieves an average consensus under the consensus protocol (2).

Proof. We can label the edges of $\mathcal{G}$ such that the $n - 1$ edges in $\mathcal{T}$ are $e_1, e_2, \dots, e_{n-1}$ and the $m - n + 1$ edges in $\mathcal{E} \setminus \mathcal{E}(\mathcal{T})$ are $e_n, e_{n+1}, \dots, e_m$. The incidence matrix corresponding to this labeling can be written as $\mathbf{H} = [\mathbf{H}_{\mathcal{E}(\mathcal{T})}^T, \mathbf{H}_{\mathcal{E} \setminus \mathcal{E}(\mathcal{T})}^T]^T$, where $\mathbf{H}_{\mathcal{E}(\mathcal{T})} \in \mathbb{R}^{(n-1) \times n}$ represents $n - 1$ edges of $\mathcal{T}$ and $\mathbf{H}_{\mathcal{E} \setminus \mathcal{E}(\mathcal{T})} \in \mathbb{R}^{(m-n+1) \times n}$ represents the remaining edges in the graph. Note that the rows of $\mathbf{H}_{\mathcal{E} \setminus \mathcal{E}(\mathcal{T})}$ are linearly dependent on the rows of $\mathbf{H}_{\mathcal{E}(\mathcal{T})}$ (Zelazo & Mesbahi, 2011). Specifically, there exists $\mathbf{T} \in \mathbb{R}^{(m-n+1) \times (n-1)}$ such that $\mathbf{T} \mathbf{H}_{\mathcal{E}(\mathcal{T})} = \mathbf{H}_{\mathcal{E} \setminus \mathcal{E}(\mathcal{T})}$. Any equilibrium point of (3) satisfies $\dot{\mathbf{x}} = -\bar{\mathbf{H}}^T \text{blkdiag}(\mathbf{A}_k) \bar{\mathbf{H}} \mathbf{x} = \mathbf{0}$. Thus, $\mathbf{x}^T \bar{\mathbf{H}}^T \text{blkdiag}(\mathbf{A}_k) \bar{\mathbf{H}} \mathbf{x} = 0$, which in turn implies $\|\text{blkdiag}(\mathbf{A}_k^{1/2}) \bar{\mathbf{H}} \mathbf{x}\|^2 = \|\mathbf{M} \mathbf{x}\|^2 = 0$. Denoting $\bar{\mathbf{T}} = \mathbf{T} \otimes \mathbf{I}_d$, this condition is equivalent to

$$\mathbf{M} \mathbf{x} = \begin{bmatrix} \text{blkdiag}(\mathbf{A}_k^{1/2})_{k=1}^{n-1} \bar{\mathbf{H}}_{\mathcal{E}(\mathcal{T})} \mathbf{x} \\ \text{blkdiag}(\mathbf{A}_k^{1/2})_{k=n}^m \bar{\mathbf{T}} \bar{\mathbf{H}}_{\mathcal{E}(\mathcal{T})} \mathbf{x} \end{bmatrix} = \mathbf{0}. \quad (6)$$

Observe that $\text{blkdiag}(\mathbf{A}_k^{1/2}) \bar{\mathbf{H}}_{\mathcal{E}(\mathcal{T})} \mathbf{x} = \mathbf{0}$ is equivalent to $\bar{\mathbf{H}}_{\mathcal{E}(\mathcal{T})} \mathbf{x} = \mathbf{0}$. Since $\mathbf{H}_{\mathcal{E}(\mathcal{T})}$ is the incidence matrix corresponding to a tree, $\mathcal{N}(\mathbf{H}_{\mathcal{E}(\mathcal{T})}) = \text{span}\{\mathbf{1}_n\}$, which means $\mathcal{N}(\bar{\mathbf{H}}_{\mathcal{E}(\mathcal{T})}) = \mathcal{R}$. Based on Theorem 2, the equilibrium $\mathbf{x}^* = \mathbf{1}_n \otimes \bar{\mathbf{x}}$ is unique and globally exponentially stable. ■

Lemma 6. Suppose there exists a positive tree $\mathcal{T} \subset \mathcal{G}$ of l vertices, under consensus protocol (2), $\mathbf{x}_i(t) \rightarrow \mathbf{x}_j(t), \forall i, j \in \mathcal{T}$, as $t \rightarrow \infty$.

Proof. By writing $\mathbf{x} = [\mathbf{x}_{\mathcal{T}}^T, \mathbf{x}_{\mathcal{V} \setminus \mathcal{V}(\mathcal{T})}^T]^T$, we express $\mathbf{H} = \begin{bmatrix} \mathbf{H}_1 & \mathbf{0} \\ \mathbf{H}_2 & \mathbf{0} \\ \mathbf{H}_3 & \mathbf{H}_4 \end{bmatrix}$, where $[\mathbf{H}_1 \ \mathbf{0}] \in \mathbb{R}^{(l-1) \times n}$ associates with the edges belonging to the tree $\mathcal{T}$, $[\mathbf{H}_2 \ \mathbf{0}]$ associates with the other edges between vertices in $\mathcal{V}(\mathcal{T})$ which do not belong to the tree, and $[\mathbf{H}_3 \ \mathbf{H}_4]$ associates with the remaining edges in $\mathcal{E}$. Observe that $\mathbf{H}_2$ is linearly dependent on $\mathbf{H}_1$ and this dependency is characterized by $\mathbf{H}_2 = \mathbf{T} \mathbf{H}_1$. The equilibria of (3) must satisfy $\bar{\mathbf{H}} \mathbf{x} = \mathbf{0}$, which implies that

$$\text{blkdiag}(\mathbf{A}_k^{1/2})_{k=1}^{l-1} \bar{\mathbf{H}}_1 \mathbf{x}_{\mathcal{T}} = \mathbf{0}. \quad (7)$$

Since $\mathcal{T}$ is a positive tree, $\bar{\mathbf{H}}_1 \mathbf{x}_{\mathcal{T}} = \mathbf{0}$. Thus, any equilibrium $\mathbf{x}^*$ of (3) must have $\mathbf{x}_{\mathcal{T}}^* \in \mathcal{N}(\bar{\mathbf{H}}_1) = \text{Range}\{\mathbf{I}_l \otimes \mathbf{I}_d\}$, i.e., all states of l agents belonging to the positive tree $\mathcal{T}$ are the same. On the basis of Theorem 1, all agents in $\mathcal{T}$ asymptotically reach a consensus. ■

The following result provides a condition to determine whether or not two vertices belong to a same cluster.

Theorem 3. Given a positive tree $\mathcal{T}$, let the cluster $\mathcal{C}(\mathcal{T})$ generated from $\mathcal{T}$ be containing:

- (i) all vertices in $\mathcal{T}$,
- (ii) any vertex $i \notin \mathcal{T}$, which defines the set $S_i = \{\mathcal{P}_k = \{v_1^k \dots v_{|\mathcal{P}_k|}^k\} | v_1^k = i, v_{|\mathcal{P}_k|}^k \in \mathcal{T}, \text{ and } \forall j = 1, \dots, |\mathcal{P}_k| - 1, v_j^k \notin \mathcal{T}\}$, satisfying

(a) For each path $\mathcal{P}_k$, let $\mathcal{N}(\mathcal{P}_k) = \bigcup_{j=1}^{|\mathcal{P}_k|-1} \mathcal{N}(\mathbf{A}_{v_j^k v_{j+1}^k})$, it holds

$$\dim \left(\bigcap_{k=1}^{|\mathcal{S}_i|} \mathcal{N}(\mathcal{P}_k) \right) = 0. \quad (8)$$

(b) Each $\mathcal{P}_k \in \mathcal{S}_i$ has no loop, i.e. $v_l \neq v_m, \forall v_l, v_m \in \mathcal{P}_k$.

Then, under the consensus protocol (2), all agents in the cluster $\mathcal{C}(\mathcal{T})$ have the same equilibrium state. Furthermore, in algorithmic perspective, the set $\mathcal{S}_i$ is finite.

Proof. From Lemma 6, all $\mathbf{x}_j$ ($j \in \mathcal{V}(\mathcal{T})$) converge to a common value $\mathbf{x}_{\mathcal{T}}^*$. Consider a vertex $i \notin \mathcal{V}(\mathcal{T})$ satisfying the condition (ii). Let $\mathbf{x}_i^*$ be the equilibrium state of agent i . Then, according to the definition of $\mathcal{N}(\mathcal{P}_k)$, there holds

$$\mathbf{x}_i^* - \mathbf{x}_{\mathcal{T}}^* \in \mathcal{N}(\mathcal{P}_k), \forall \mathcal{P}_k \in \mathcal{S}_i. \quad (9)$$

It follows from (8) that the only solution for (9) is $\mathbf{x}_i^* - \mathbf{x}_{\mathcal{T}}^* = \mathbf{0}$, or $\mathbf{x}_i^* = \mathbf{x}_{\mathcal{T}}^*$. Thus, the cluster $\mathcal{C}(\mathcal{T})$ reaches a consensus. Suppose $\mathcal{P}_1$ and $\mathcal{P}_2$ are two paths, and $\mathcal{P}_2$ is obtained by adding loops to $\mathcal{P}_1$. Then, $\mathcal{N}(\mathcal{P}_1) \subseteq \mathcal{N}(\mathcal{P}_2)$. It follows that $\mathcal{N}(\mathcal{P}_1) \cap \mathcal{N}(\mathcal{P}_2) = \mathcal{N}(\mathcal{P}_1)$. Thus, it is unnecessary to consider loops when checking condition (8), which means that $|\mathcal{S}_i|$ is finite. Finally, consider a vertex i which does not satisfy both (i) and (ii). Let $\mathcal{S}_i$ be the set of paths from i to $\mathcal{T}$ with $\dim(\bigcap_{k=1}^{|\mathcal{S}_i|} \mathcal{N}(\mathcal{P}_k)) \geq 1$. Clearly, there exists nontrivial solution satisfying $\mathbf{x}_i^* - \mathbf{x}_{\mathcal{T}}^* \neq \mathbf{0}$. ■

Corollary 2. Consider two positive trees $\mathcal{T}_1, \mathcal{T}_2$ connected via the edge set $\mathcal{S} = \{(i, j) | i \in \mathcal{T}_1, j \in \mathcal{T}_2\}$. If $\sum_{(i,j) \in \mathcal{S}} \mathbf{A}_{ij}$ is positive definite, the equilibrium states of all agents in $\mathcal{T}_1$ and $\mathcal{T}_2$ are the same under the consensus protocol (2).

Lemma 7 (Partitioning a Graph into Positive Trees). Given a positive semiconnected graph $\mathcal{G}$, consider a set of positive trees $\{\mathcal{T}_1, \dots, \mathcal{T}_p\}$ ($1 \leq p \leq n$) satisfying

- (i) $\mathcal{V}(\mathcal{T}_m) \cap \mathcal{V}(\mathcal{T}_l) = \emptyset, \bigcup_{m=1}^p \mathcal{V}(\mathcal{T}_m) = \mathcal{V}(\mathcal{G})$,
- (ii) For each $\mathcal{T}_k$ ($1 \leq k \leq p$), $i, j \in \mathcal{V}(\mathcal{T}_k)$ if and only if there exists a positive path from i to j .

Then, the partition $\{\mathcal{V}(\mathcal{T}_1), \dots, \mathcal{V}(\mathcal{T}_p)\}$ of $\mathcal{G}$ is unique.

Algorithm 1 Finding clusters of a matrix-weighted graph $\mathcal{G}$

Require: $\mathcal{G}(\mathcal{V}, \mathcal{E}, \mathbf{A})$

```

1:  $i \leftarrow 0$ ;
2: Find the set of positive trees  $\{\mathcal{T}_1, \dots, \mathcal{T}_p\}$  in  $\mathcal{G}$ ;
3:  $\mathcal{C}_{\mathcal{G}}(0) \leftarrow \{\mathcal{C}_m = \{\mathcal{V}(\mathcal{T}_m)\}, m = 1, \dots, p\}$ ;
4: repeat
5:    $\mathcal{C}_{\mathcal{G}}(i+1) \leftarrow \mathcal{C}_{\mathcal{G}}(i)$ ;
6:   check  $\leftarrow$  false;
7:   for all  $\mathcal{C}_m \in \mathcal{C}_{\mathcal{G}}(i)$  do
8:     for all  $\mathcal{C}_l \in \mathcal{C}_{\mathcal{G}}(i), l \neq m$  do
9:       if  $\exists$  vertex  $k \in \mathcal{C}_l$  satisfies Theorem 3(ii) then
10:         $\mathcal{V}_{temp} \leftarrow \mathcal{V}(\mathcal{T}_m) \cup \mathcal{V}(\mathcal{T}_l)$ ;
11:         $\mathcal{E}_{temp} \leftarrow \mathcal{E}(\mathcal{T}_m) \cup \mathcal{E}(\mathcal{T}_l) \cup \mathcal{S}$ ;
12:         $\mathcal{C}_{temp} \leftarrow \mathcal{C}_m \cup \mathcal{C}_l$ ;
13:         $\mathcal{C}_{\mathcal{G}}(i+1) \leftarrow (\mathcal{C}_{\mathcal{G}}(i+1) \setminus \{\mathcal{C}_m, \mathcal{C}_l\}) \cup \{\mathcal{C}_{temp}\}$ ;
14:        check  $\leftarrow$  true;
15:        break;
16:      end if
17:    end for
18:    if check == true then
19:      break;
20:    end if
21:  end for
22:   $i \leftarrow i+1$ ;
23: until  $\mathcal{C}_{\mathcal{G}}(i) == \mathcal{C}_{\mathcal{G}}(i-1)$ 
```

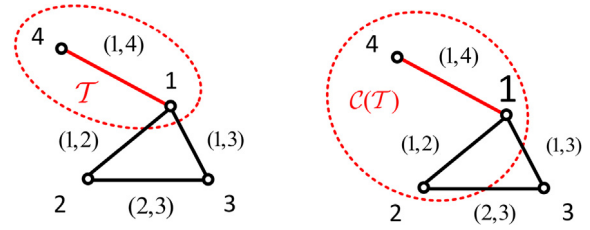


Fig. 2. Illustration of the four-agent system in Example 1.

Let $\mathcal{C}(\mathcal{T}_m)$ be the cluster generated from the positive tree $\mathcal{T}_m$. If there exists a vertex $i \in \mathcal{C}(\mathcal{T}_m)$ satisfying the condition (ii) in Theorem 3 with a cluster $\mathcal{C}(\mathcal{T}_l)$, we can form a new cluster $\mathcal{C}(\mathcal{T}_m) \cup \mathcal{C}(\mathcal{T}_l)$ by merging $\mathcal{C}(\mathcal{T}_m)$ and $\mathcal{C}(\mathcal{T}_l)$. By this procedure, we can extend the positive trees in the graph. All vertices in the new cluster reach a consensus under (2). To check whether two clusters $\mathcal{C}(\mathcal{T}_m)$ and $\mathcal{C}(\mathcal{T}_l)$ can be merged or not, it is sufficient to check condition (ii) of Theorem 3 for only one vertex $i \in \mathcal{C}(\mathcal{T}_m)$ with regard to $\mathcal{C}(\mathcal{T}_l)$. Algorithm 1 provides a method for finding all clusters in the graph by iteratively checking condition (ii) in Theorem 3 and merging clusters together. Initially, there are p clusters $\mathcal{C}_m$ ($m = 1, \dots, p$) containing only vertices of $\mathcal{T}_m$. After each iteration, a cluster may remain or be merged into another cluster. The number of clusters at each step i could be reduced by 1 or equal to which of step $i-1$. Thus, Algorithm 1 must terminate after finite iterations and the output is a set of clusters $\mathcal{C}_{\mathcal{G}} = \{\mathcal{C}_1, \dots, \mathcal{C}_q\}$ ($1 \leq q \leq p$) satisfying: (i) $\mathcal{C}_m \cap \mathcal{C}_l = \emptyset, \bigcup_{m=1}^q \mathcal{C}_m = \mathcal{V}(\mathcal{G})$ and (ii) For $1 \leq m \neq l \leq q$, $\mathcal{C}_m$ and $\mathcal{C}_l$ cannot be merged together. We can now state the following theorem.

Theorem 4. Under the consensus protocol (2), an average consensus is achieved if and only if $\mathcal{G}$ is spanned by a cluster.

Proof. If $\mathcal{G}$ is spanned by a cluster, based on Theorem 3 the equilibrium state of all agents in the graph is the same, i.e. $\mathbf{x}^*$ is the only equilibrium point of (3). Thus, the average consensus is globally asymptotically achieved based on Theorem 1. However, if there is no cluster spanning $\mathcal{G}$, the agents belonging to two clusters of $\mathcal{G}$ may not agree and an average consensus cannot be globally achieved. ■

Example 1. Consider a 4-agent system in $\mathbb{R}^3$ with the interaction graph depicted in Fig. 2 and the matrix weights: $\mathbf{A}_{12} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{A}_{13} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \mathbf{A}_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{A}_{14} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. There is a positive tree $\mathcal{T}$ containing vertices 1 and 4. Further, we have $\mathcal{N}(\mathbf{A}_{12}) = \text{span}\{[1, 0, 0]^T\}$, $\mathcal{N}(\mathbf{A}_{13}) = \text{span}\{[0, 1, 0]^T, [0, 0, 1]^T\}$, and $\mathcal{N}(\mathbf{A}_{23}) = \text{span}\{[0, 1, 0]^T\}$. There are two paths from vertex 2 to vertex 1: $\mathcal{P}_1 = 21$, and $\mathcal{P}_2 = 231$, where $\mathcal{N}(\mathcal{P}_1) = \mathcal{N}(\mathbf{A}_{12}) = \text{span}\{[1, 0, 0]^T\}$, and $\mathcal{N}(\mathcal{P}_2) = \mathcal{N}(\mathbf{A}_{13}) \cup \mathcal{N}(\mathbf{A}_{23}) = \text{span}\{[0, 1, 0]^T, [0, 0, 1]^T\}$. It follows $\mathcal{N}(\mathcal{P}_1) \cap \mathcal{N}(\mathcal{P}_2) = \{\mathbf{0}\}$, which implies that agent 2 is in $\mathcal{C}(\mathcal{T})$ due to Theorem 3(ii). Consider the vertex 3, which has two paths to the cluster $\mathcal{C} = \{1, 2, 4\}$: $\mathcal{P}_3 = 31$ and $\mathcal{P}_4 = 32$. Since $\mathcal{N}(\mathcal{P}_3) \cap \mathcal{N}(\mathcal{P}_4) = \mathcal{N}(\mathbf{A}_{23}) = \text{span}\{[0, 1, 0]^T\}$, vertex 3 does not belong to $\mathcal{C}$. From numerical simulation, as depicted in Fig. 3, we have $\mathbf{x}_1^* = \mathbf{x}_2^* = \mathbf{x}_4^* \neq \mathbf{x}_3^*$, which is as expected from the above discussion.

The next result follows from Theorems 2 and 4.

Theorem 5. Given a matrix-weighted graph $\mathcal{G}$, there is a cluster spanning all vertices of $\mathcal{G}$ if and only if $\mathcal{N}(\mathbf{L}) = \mathcal{R}$.

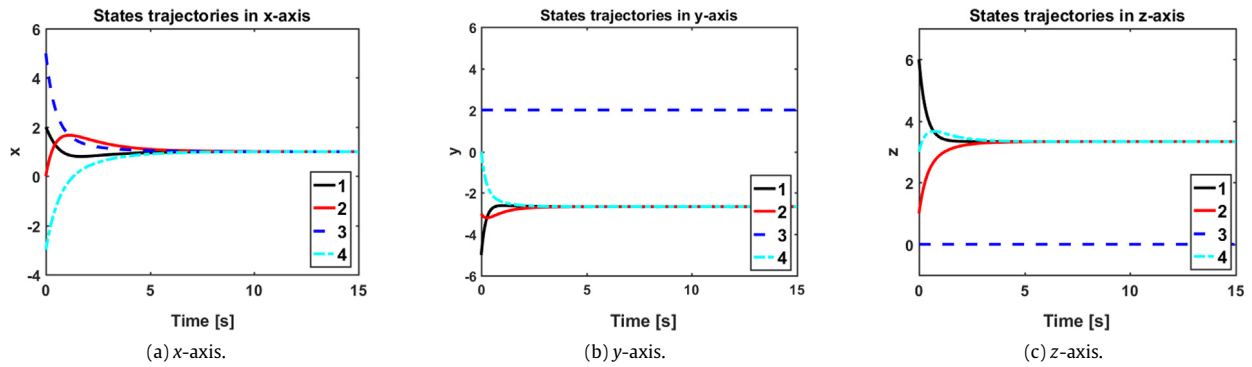


Fig. 3. The states along 3 axes of four agents under (2).

5. Applications

The matrix-weighted consensus proposed in this paper can be applied to various networked systems. For examples, it can be utilized to formulate distributed formation control in multi-agent systems. Also, as mentioned in introduction, it is natural to model the logical inter-dependency of opinions of social agents or states of biological agents by matrix-weighted edges. It is noticeable that uncertainties of inter-dependency between agents or errors in sensing/control variables in distributed agents can be handled by matrix-weighted edge sets. Due to the space limitation, we omit the detailed applications. For more technical details, it is recommended to refer to Trinh and Ahn (2017) and Trinh, Ye, Ahn, and Anderson (2017).

6. Conclusion

In this paper, the matrix-weighted consensus algorithm was proposed. Due to the existence of semipositive definite connections, cluster phenomenon may easily happen in the network. A global average consensus can be achieved if and only if the nullspace of the matrix-weighted Laplacian is spanned by the consensus space $\mathcal{R}$, or equivalently, the graph is spanned by a cluster. Also, an algorithm for finding all clusters in a given matrix-weighted graph was provided. For future works, we would like to investigate a consensus problem of directed or switching/random networks with matrix weights.

Acknowledgments

Due to the space limitation, we refer readers to the technical report (Trinh & Ahn, 2017) for several omitted proofs and

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